190 MeV/u Polarized Deuteron on ^{124}Sn Local Full-Chain Rate Estimate

Only locally reproducible ImQMD + Geant4 results are retained

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Scope

- This English report keeps only **locally reproducible** results.
- All headline numbers come from the current local chain:
  - ImQMD raw breakup sample in `z_pol/b_discrete`,
  - fresh Geant4 rerun in the current detector geometry,
  - rate conversion using the recomputed ^{124}Sn number density.
- Results quoted from `polarization.tex` are intentionally removed from the main result tables.
- The present quantitative scope is:
  - target: ^{124}Sn,
  - beam energy: 190 MeV/u,
  - sample: z polarization, $\gamma = 0.60$,
  - impact-parameter range used in the shell sum: $b = 1 \ldots 9$ fm.

Inputs Used in the Local Calculation

- Beam intensity: $I = 1.0 \times 10^7$ pps
- Beam time for the headline yield: $t = 16$ h
- Beam energy per nucleon: $T_u = 190$ MeV/u
- SRC harmonic number: $h = 6$
- SRC extraction radius: $R_{\mathrm{ext}} = 5.36$ m
- Tin density used for target conversion: $\rho = 7.31\ \mathrm{g/cm^3}$
- Enriched-^{124}Sn number density:

$$n = \frac{N_A \rho}{M_{124}} = \frac{(6.022 \times 10^{23})(7.31)}{124} = 3.5501 \times 10^{22}\ \mathrm{cm}^{-3}$$

The local full-chain quantities taken from `validation_summary.json` are:

- selected-input cross section: $\sigma_{\mathrm{sel}} = 516.5$ mb
- detected-coincidence cross section: $\sigma_{\mathrm{coin}} = 204.2$ mb

How the Bunch Interval Is Calculated

$$\gamma = 1 + \frac{T_u}{m_u c^2} = 1 + \frac{190}{931.494} = 1.20397,$$

$$\beta = \sqrt{1 - \gamma^{-2}} = 0.55689,$$

$$v = \beta c = 1.6695 \times 10^8 \text{ m/s},$$

$$C = 2\pi R_{\text{ext}} = 2\pi \times 5.36 = 33.678 \text{ m},$$

$$f_{\text{rev}} = \frac{v}{C} = 4.9573 \text{ MHz},$$

$$f_{\text{bucket}} = h f_{\text{rev}} = 29.7441 \text{ MHz},$$

$$\Delta t = \frac{1}{f_{\text{bucket}}} = 33.62 \text{ ns}.$$

$$\mu_{\text{beam/bucket}} = \frac{I}{f_{\text{bucket}}} \approx 0.3362$$

So the bunch interval is obtained from beam kinematics and SRC machine parameters, not from an assumed timing constant.

How the Production and Coincidence Cross Sections Are Calculated

Step 1: ImQMD breakup probability in each shell

$$P_{\mathrm{bu}}(b_i) = \frac{N_{\mathrm{breakup}}(b_i)}{N_{\mathrm{incident}}(b_i)}$$

Step 2: shell-sum cross section for the current `b_discrete` dataset

$$\sigma \approx \sum_i 2\pi b_i \Delta b \, P(b_i)$$

Step 3: propagate the selected events through Geant4

$$P_{\mathrm{sel}}(b_i) = \frac{N_{\mathrm{G4}}(b_i)}{N_{\mathrm{incident}}(b_i)}, \qquad P_{\mathrm{coin}}(b_i) = \frac{N_{\mathrm{coin}}(b_i)}{N_{\mathrm{incident}}(b_i)}$$

$$\sigma_{\mathrm{sel}} \approx \sum_i 2\pi b_i \Delta b \, P_{\mathrm{sel}}(b_i), \qquad \sigma_{\mathrm{coin}} \approx \sum_i 2\pi b_i \Delta b \, P_{\mathrm{coin}}(b_i)$$

Important interpretation

- The main coincidence rate in this report is **not** obtained from a geometry-only solid-angle factor.
- The detector geometry is defined by the current Geant4 macro, but the quoted coincidence rate comes from **Geant4 hit counting** in that geometry.
- In the current validation script, a coincidence event means the same event has both:
  - at least one PDC fragment hit,
  - at least one NEBULA hit.

Current Local Full-Chain Statistics

Quantity	Value	Comment
Raw-file header incident events	180000	18 files, 10000 events each
Stored breakup rows before cut	28611	breakup events present in raw files
Events propagated through Geant4	22174	after generator selection and cut
Fresh coincidence hits	8408	Geant4 rerun result
Fresh coincidence efficiency on propagated sample	37.92%	8408/22174
Archive coincidence efficiency	37.95%	archived rerun for comparison
Fresh minus archive	−0.027 points	reproducibility check

The Geant4 coincidence efficiency is stable at the level of ~ 0.03 percentage points between the fresh rerun and the archived output.

Cross Sections Obtained from the Local Chain

Quantity	Value	Meaning
Raw breakup cross section	678.2 mb	ImQMD breakup shell sum
Selected-input cross section	516.5 mb	events that enter the current Geant4 chain
Detected-coincidence cross section	204.2 mb	events counted as PDC+NEBULA coincidence
Naive $\pi b_{\max}^2 (N/N_0)$ estimate	404.5 mb	shown only as a comparison

- The recommended result for the present `b_discrete` data structure is the shell-sum estimate, not the naive area formula.
- For rate conversion in this report:

$$R_{\mathrm{sel}} = ndl\sigma_{\mathrm{sel}}, \qquad R_{\mathrm{coin}} = ndl\sigma_{\mathrm{coin}}$$

- Here σ_{coin} already includes the current Geant4 detector acceptance and coincidence definition.

Thickness Scan: Local Full-Chain Results for ^{124}Sn

Thickness (mm)	R_{sel} (kHz)	R_{coin} (kHz)	16 h coincidence yield	$\mu_{\mathrm{coin/bucket}}$
0.5	9.169	3.625	2.09×10^{8}	1.22×10^{-4}
1.0	18.338	7.250	4.18×10^{8}	2.44×10^{-4}
1.5	27.506	10.875	6.26×10^{8}	3.66×10^{-4}
2.0	36.675	14.500	8.35×10^{8}	4.87×10^{-4}
3.0	55.013	21.750	1.25×10^{9}	7.31×10^{-4}

- Even with the local full-chain coincidence cross section, the single-bucket coincidence occupancy remains well below 10^{-3}.
- Under the current local chain, the 3 mm reference gives:
  - $R_{\mathrm{coin}} \approx 21.75$ kHz
  - $N_{\mathrm{coin}}(16\ \mathrm{h}) \approx 1.25 \times 10^{9}$

Inventory-Constrained Disk Designs

Disk diameter (mm)	Areal density (mg/cm^2)	Bulk thickness (mm)	R_{coin} (kHz)	16 h coincidence yield
12	1061	1.451	10.523	6.06×10^8
15	679	0.929	6.735	3.88×10^8
20	382	0.523	3.788	2.18×10^8

- All three options assume the same inventory constraint: 1.2 g of tin per isotope.
- The 15 mm disk remains a practical compromise:
  - wider alignment tolerance than 12 mm,
  - still much higher rate than 20 mm.

Where the Coincidence Rate Comes From

- The coincidence rate quoted in this report comes from the **full chain**:

$$\mathrm{ImQMD} \rightarrow \text{event selection} \rightarrow \mathrm{Geant4} \rightarrow \mathrm{PDC} + \mathrm{NEBULA}\ \text{coincidence counting}$$

- Therefore:
  - the breakup production part comes from ImQMD shell-by-shell probabilities,
  - the detector part comes from Geant4 transport through the current detector geometry,
  - the final coincidence rate uses the detected-coincidence cross section $\sigma_{\mathrm{coin}} = 204.2$ mb.
- What is **not** used as a headline result here:
  - `polarization.tex` rate coefficients,
  - geometry-only acceptance proxies,
  - target placeholders without a same-chain extraction.

Headline Numbers from the Current Local Chain

- Bunch spacing at 190 MeV/u in the current SRC setting:

$$\Delta t = 33.62\ \mathrm{ns}$$

- For a 3 mm ^{124}Sn reference target:

$$R_{\mathrm{coin}} \approx 21.75\ \mathrm{kHz}, \qquad N_{\mathrm{coin}}(16\ \mathrm{h}) \approx 1.25 \times 10^{9}$$

- For the 15 mm diameter, 1.2 g inventory-constrained disk:

$$d \approx 0.929\ \mathrm{mm}, \qquad R_{\mathrm{coin}} \approx 6.735\ \mathrm{kHz}, \qquad N_{\mathrm{coin}}(16\ \mathrm{h}) \approx 3.88 \times 10^{8}$$

- These numbers are local full-chain results for the current ^{124}Sn zpol $\gamma = 0.60$ sample only.